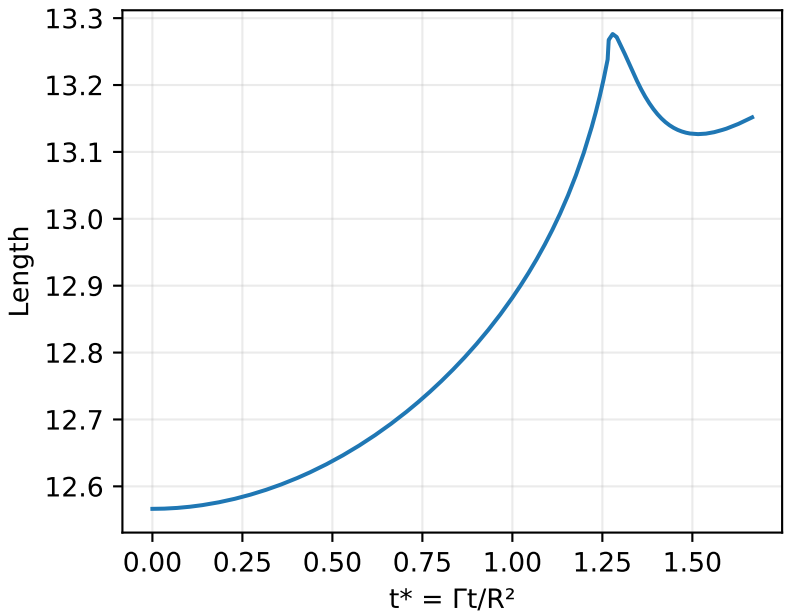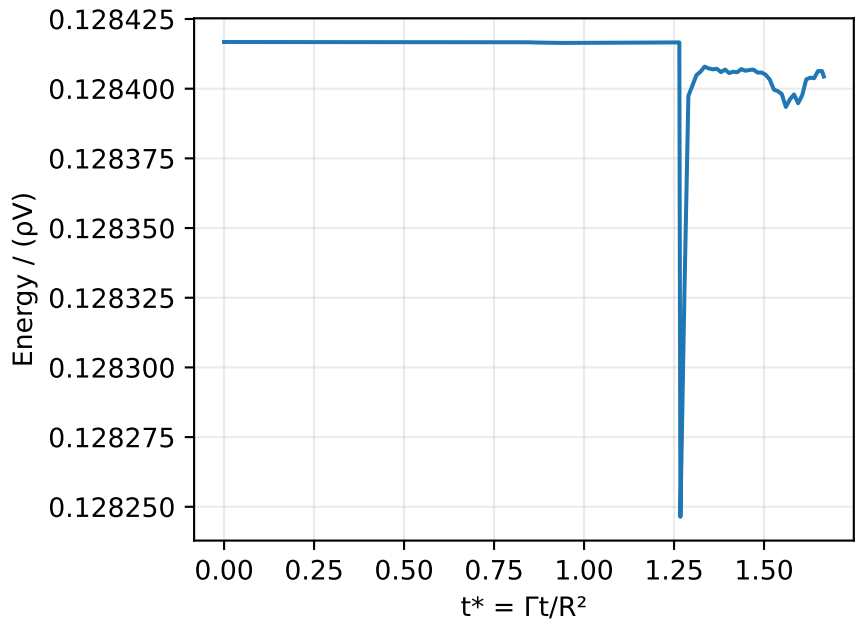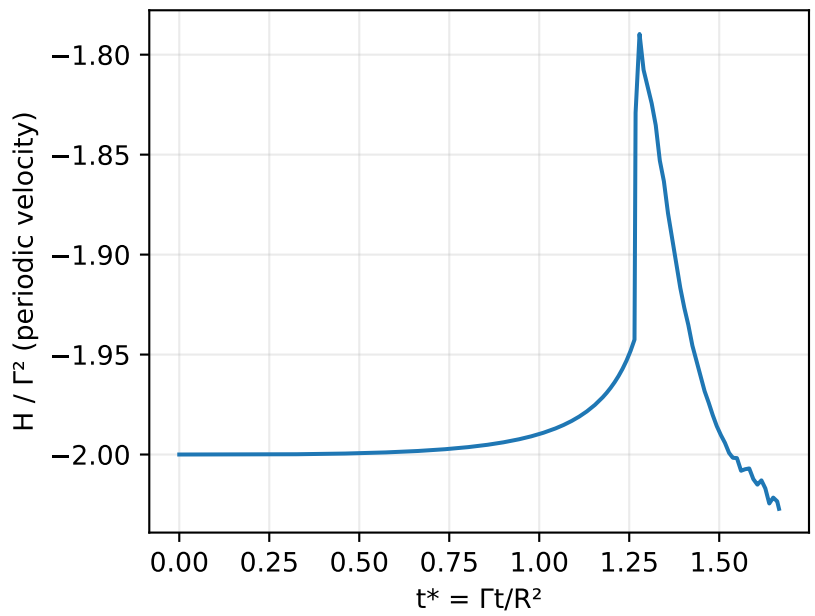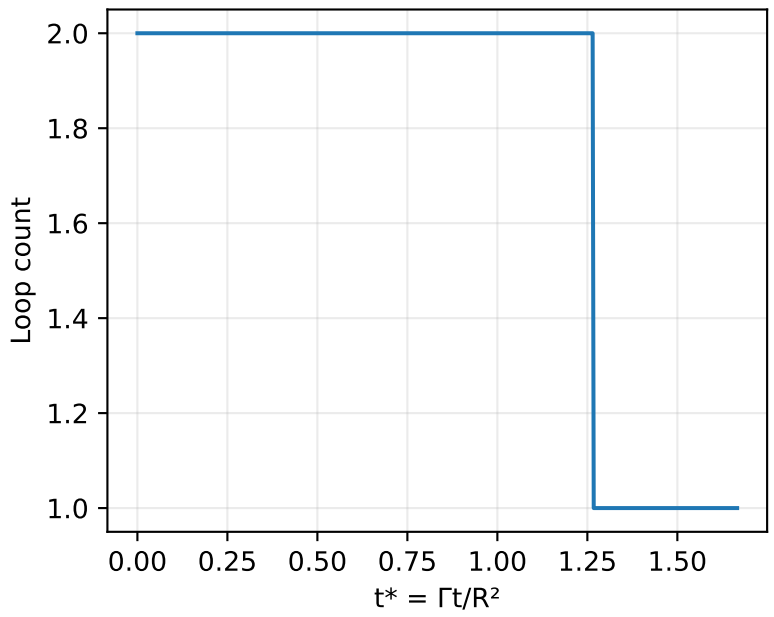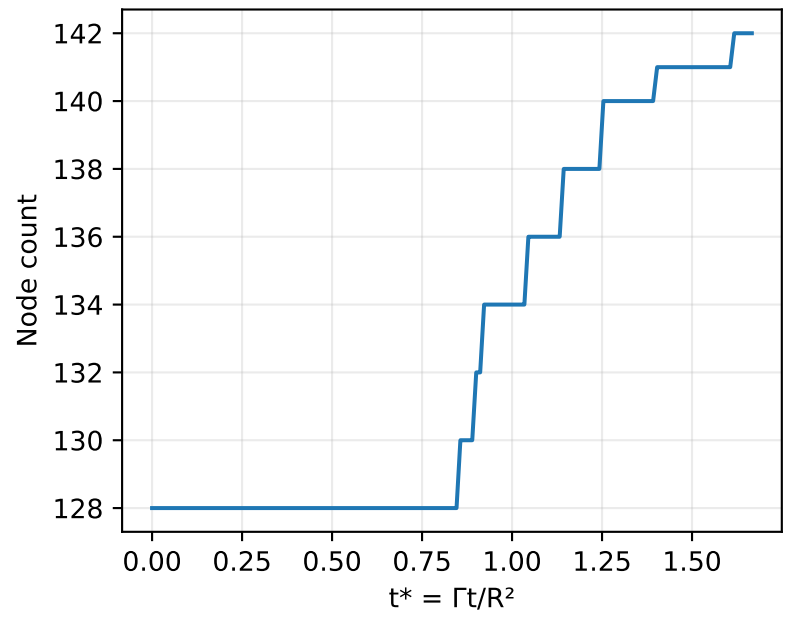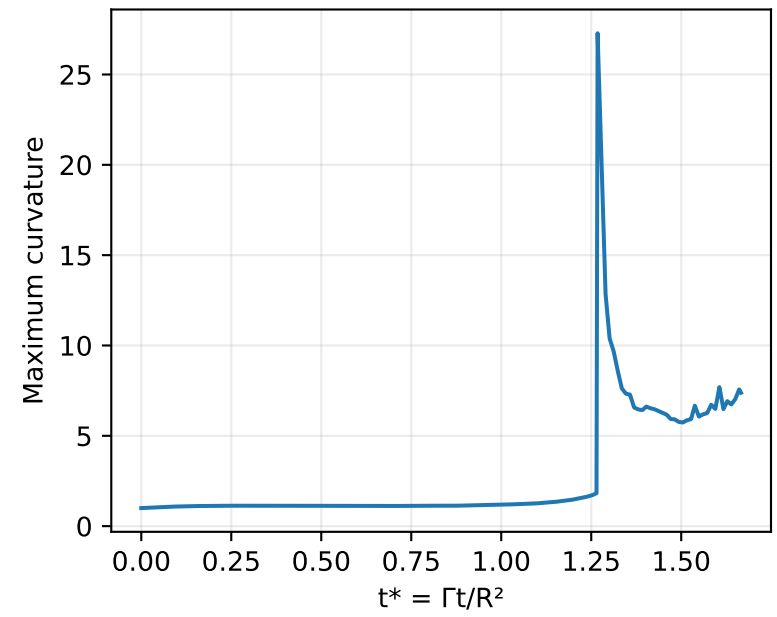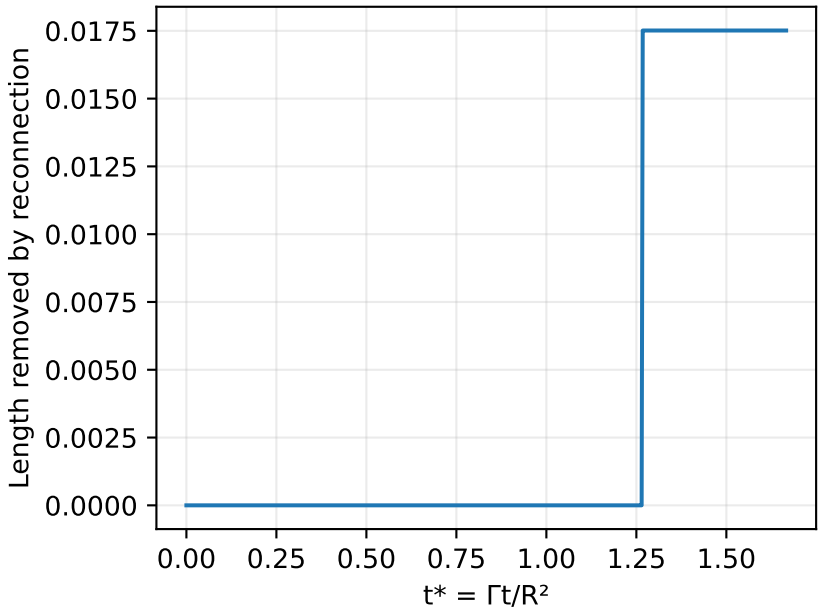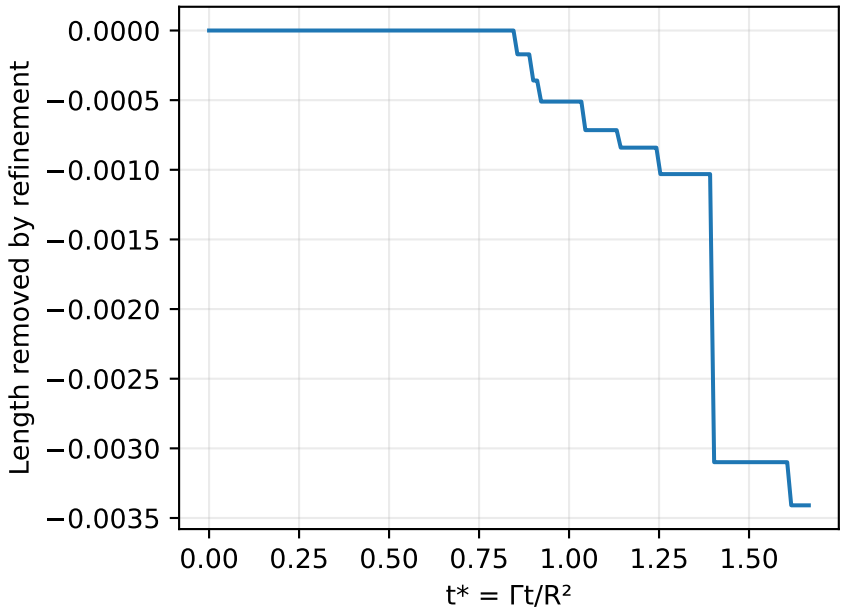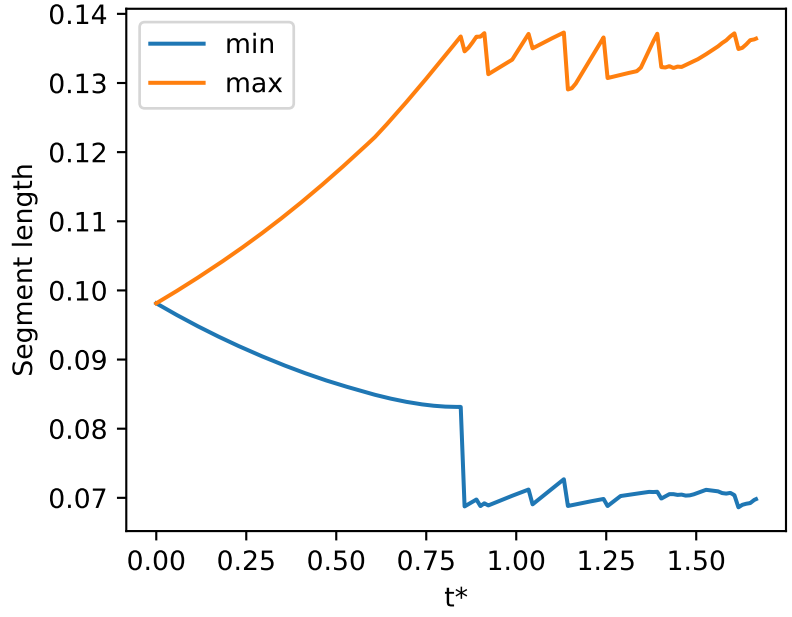hopf_d100 · a=0.001 · no mutual friction